An Ablation Study of Common Training Techniques for CNNs on CIFAR-10

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Abstract

This report dives into common techniques for improving the performance of a small convolutional neural network (CNN) trained on CIFAR-10 and constrained by compute resources. A baseline CNN was compared against controlled variants using data augmentation, batch normalization, dropout, label smoothing, and all techniques combined. Because of the nature of this experiment, an ablation study was conducted that changed one part of the baseline pipeline while keeping the dataset, architecture, optimizer, learning rate, batch size, number of epochs, and random seed fixed. The strongest single technique was data augmentation, which improved test accuracy from 73.61% to 81.63%. In comparison, the CNN model using all techniques reached 81.32% accuracy and produced the most stable test-loss behavior. However, its measured training accuracy was lower than the test accuracy because training metrics were affected by training-time augmentation and regularization.

1 Introduction

Modern deep learning models often depend on the use of training techniques that enhance a model's output beyond its architecture. A few of the most common techniques used include data augmentation, batch normalization, dropout, and label smoothing. However, the effect they have on output quality is unclear, especially when they are applied at the same time. This project aims to study those techniques in a controlled environment by training a small CNN on CIFAR-10 and adding each technique one at a time.

The research question to be addressed is: *How much do data augmentation, batch normalization, dropout, and label smoothing improve a small CNN on CIFAR-10 under limited compute?* Instead of reporting that a CNN was trained, this report provides a reproducible ablation study that measures the individual and combined contribution of each training technique.

The main result is that data augmentation produced the largest single-technique improvement, increasing test accuracy by 8.02 percentage points over the baseline. Batch normalization and dropout improved test accuracy more modestly, while label smoothing gave the smallest accuracy gain. Combining all tricks achieved strong test accuracy, but it did not exceed augmentation alone in this one-seed run.

2 Background

The dataset used, CIFAR-10, is a standard image classification dataset containing 32-by-32 color images from 10 object classes. These classes include airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck [1]. This dataset was picked because it is well documented, diverse, and small enough for a small CNN to be used to study training choices.

The model architecture used was a convolutional neural network, a network that passes a convolving kernel over the image and uses the result as learned feature information. This makes CNNs well suited for image recognition tasks because early layers can learn edges and simple textures, while deeper layers can learn more class-specific visual patterns [2].

Data augmentation artificially increases the diversity of the same training images by applying random transformations to them. In this project, augmentation was used with random cropping and horizontal flipping. These transformations help the model recognize images that are shifted, flipped, or visually similar to other classes.

Batch normalization normalizes intermediate activations during training and can make optimization more stable [3]. In CNNs, it is commonly placed after convolutional layers and before or after nonlinearities, depending on implementation.

Dropout randomly disables units during training, forcing the model to avoid relying too heavily on any one feature pathway. It is a regularization method intended to reduce overfitting [4]. Label smoothing modifies the target labels so that the correct class is not assigned a probability of 1.0. This discourages overconfident predictions and can improve generalization or calibration, even when top-1 accuracy gains are modest [5].

3 Methods

3.1 Dataset

All experiments used CIFAR-10. The model was trained on the CIFAR-10 training split and evaluated on the CIFAR-10 test split. Each image is an RGB image with shape $3 \times 32 \times 32$, and the task is 10-class classification.

3.2 Model Architecture

Due to computational constraints, a small CNN was used rather than complex architectures like ResNet. This was an intentional choice since the goal was to determine the effects of the techniques under limited compute and not to maximize performance using state-of-the-art architectures. The model consisted of a simple pattern of convolution, ReLU activation, pooling, flattening, and fully connected classification.

Table 1: Small CNN architecture used for the ablation study.

Layer	**Details**
Input	$3 \times 32 \times 32$ CIFAR-10 image
Conv1 + ReLU	3 input channels, 32 filters, 3×3 kernel
MaxPool	2×2 pooling
Conv2 + ReLU	32 to 64 filters, 3×3 kernel
MaxPool	2×2 pooling
Conv3 + ReLU	64 to 128 filters, 3×3 kernel
Flatten	Convert feature maps to vector
Fully connected + ReLU	128 hidden units
Output layer	10 logits, one per CIFAR-10 class

3.3 Training Setup

The training setup was held fixed across experiments except for the training trick being tested. The experiments used Adam optimization, cross-entropy loss, 30 training epochs, and seed 0. For the label-smoothing experiment, the standard cross-entropy objective was replaced with cross-entropy using label smoothing. For the augmentation experiment and all-tricks experiment, random crop and random horizontal flip were applied only to the training data.

Table 2: Fixed training configuration.

Setting	**Value**
Dataset	CIFAR-10
Epochs	30
Optimizer	Adam
Learning rate	0.001
Batch size	128
Seed	0
Primary metric	Test accuracy
Secondary metrics	Train accuracy, train-test gap, train loss, test loss

3.4 Evaluation Metrics

The main evaluation metric was test accuracy. The train-test generalization gap was computed as

$$\text{Gap} = \text{Train Accuracy} - \text{Test Accuracy}.$$

A large positive gap suggests overfitting has occurred because the model performs much better on the training set than the test set. A small gap indicates better generalization. When the experiment was performed with all the techniques, the gap was negative because the recorded training accuracy was measured under training-time conditions with augmentation and regularization, while test accuracy was measured in evaluation mode on unaugmented images. Therefore, the negative gap should be interpreted carefully.

4 Experiments

The experiments were designed as an ablation study. To isolate the effect of each technique, each experiment modified only one component of the baseline training pipeline while keeping the optimizer, learning rate, batch size, number of epochs, seed, and overall model size fixed.

Table 3: Experiment variants.

Experiment	**Change from baseline**
Baseline	Plain CNN; no augmentation, batch norm, dropout, or label smoothing
Augmentation	Random crop and horizontal flip on training images
BatchNorm	Batch normalization added after convolutional layers
Dropout	Dropout added before the final classifier
Label smoothing	Cross-entropy loss with smoothed targets
All tricks	Augmentation, batch normalization, dropout, and label smoothing combined

5 Results

Table 4 summarizes the obtained results from the experiment. The test accuracy and generalization gap were read directly from the test accuracy and gap bar charts. Training accuracy was reconstructed as test accuracy plus the reported gap.

Table 4: Final accuracy results after 30 epochs.

Experiment	**Train Acc. (%)**	**Test Acc. (%)**	**Gap (pp)**	**Δ vs. Baseline (pp)**
Baseline	98.69	73.61	25.08	0.00
Augmentation	84.46	81.63	2.83	+8.02
BatchNorm	98.72	76.20	22.52	+2.59
Dropout	92.85	76.31	16.54	+2.70
Label smoothing	100.00	74.94	25.06	+1.33
All tricks	72.27	81.32	-9.05	+7.71

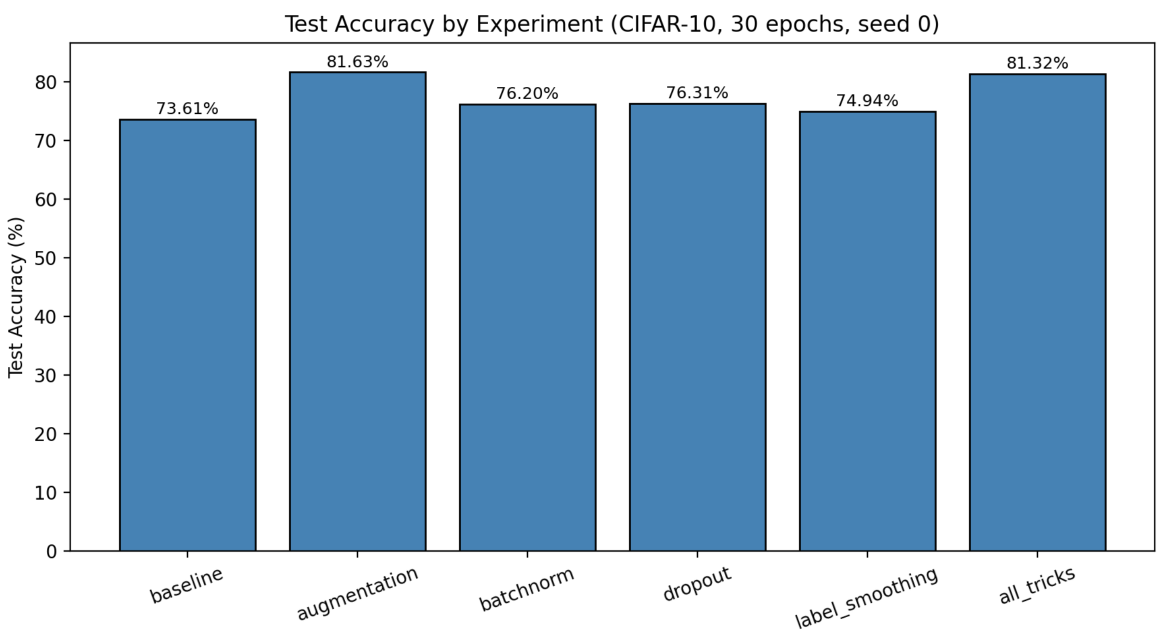

Figure 1: Final test accuracy by experiment. Data augmentation achieved the highest single-technique test accuracy at 81.63%, while all tricks combined reached 81.32%.

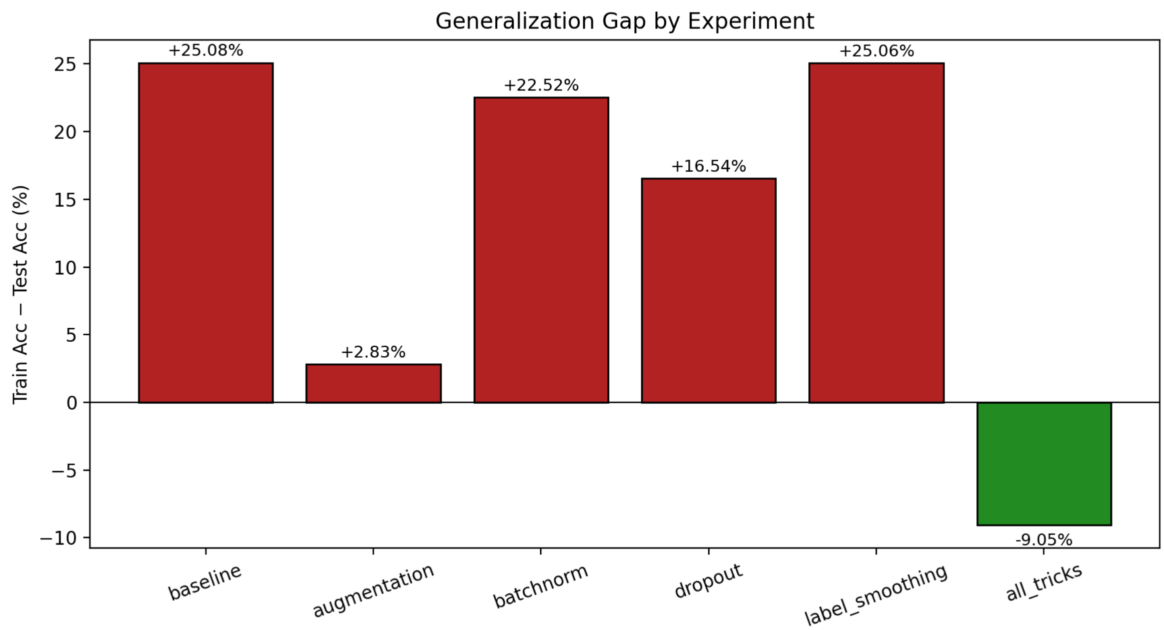

Figure 2: Train-test generalization gap by experiment. Augmentation sharply reduced the gap compared with the baseline, while the all-tricks run produced a negative gap because training accuracy was measured under stronger training-time regularization.

The baseline results were able to achieve 73.61% test accuracy; however, the training accuracy was high, at 98.69%. This suggests that overfitting occurred with the baseline model, meaning it did not generalize effectively on the test set. Data augmentation gave the biggest improvement, achieving 81.63% test accuracy. It also reduced the gap to 2.83 percentage points, suggesting better generalization. This makes sense because data augmentation exposes the model to a more diverse set of data through random cropping and flipping.

Batch normalization and dropout improved test accuracy to 76.20% and 76.31%, respectively. These were useful improvements over the baseline, but neither matched augmentation. Dropout reduced the gap more than batch normalization, suggesting that it helped regularize the model, although its accuracy gain was still modest.

Label smoothing reached 74.94% test accuracy, which was only 1.33 percentage points above the baseline. This does not mean label smoothing is useless; it may improve confidence calibration more than top-1 accuracy. However, calibration metrics such as expected calibration error were not measured in this project.

The all-tricks model achieved 81.32% test accuracy, only 0.31 percentage points below augmentation alone. Its result shows that combining techniques can produce strong performance, but it also suggests that stacking regularization methods does not automatically improve top-1 accuracy unless hyperparameters are tuned for the combined setup.

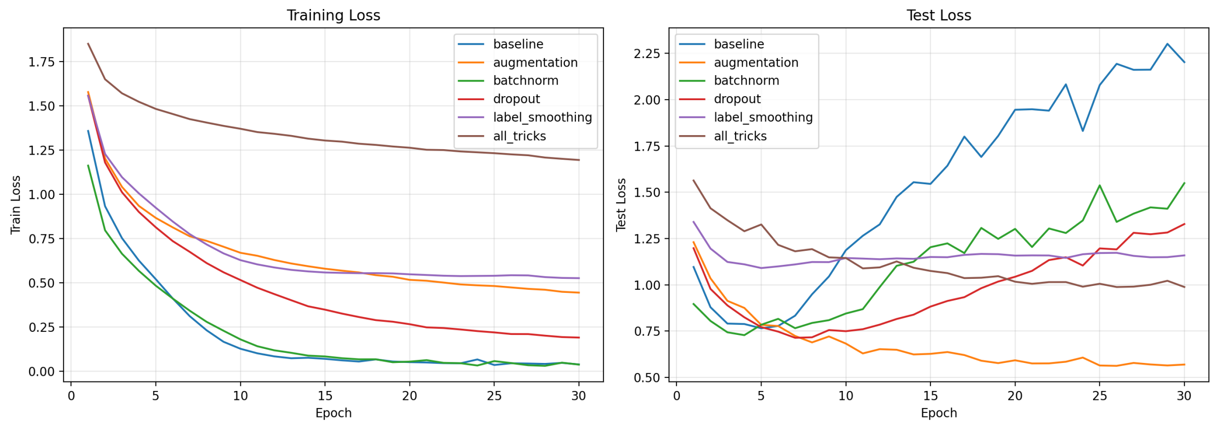

Figure 3: Training and test loss curves. The baseline training loss falls close to zero while test loss rises after the early epochs, indicating severe overfitting. Augmentation keeps the test loss lowest by the end of training.

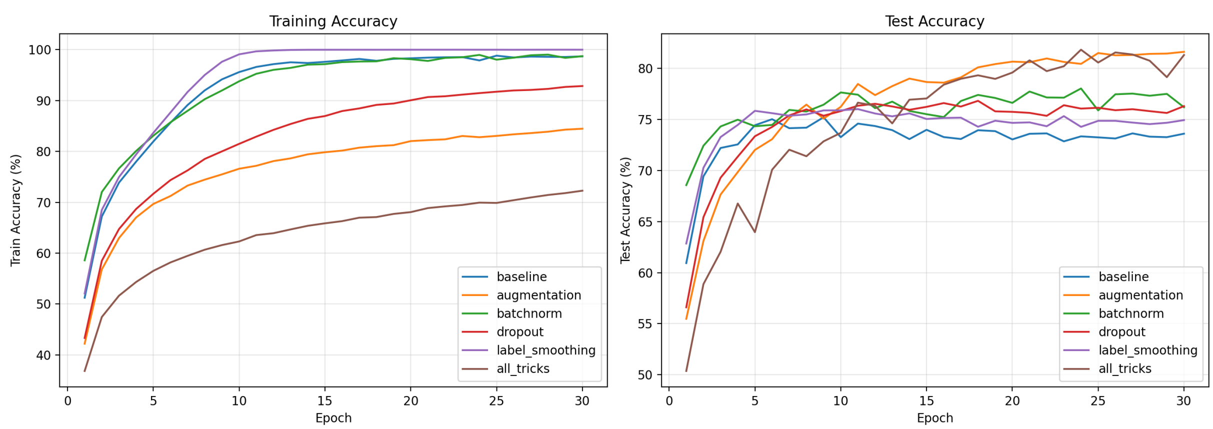

Figure 4: Training and test accuracy curves. The baseline and batch-normalized models fit the training data strongly, while augmented training improves test accuracy and narrows the generalization gap.

The loss curves provide the clearest evidence of overfitting. For the baseline, training loss decreases throughout training, while test loss begins increasing after the early epochs. This pattern means the model continues memorizing the training data after its test-set performance has stopped improving. Augmentation changes this behavior: its training accuracy is lower, but its final test accuracy and test loss are better.

The accuracy curves tell the same story. The baseline training accuracy approaches 100%, but test accuracy stays near the mid-70% range. The augmented model trains more slowly and reaches lower training accuracy, but it achieves the best final test accuracy. This is a classic regularization tradeoff: the model fits the training set less perfectly but generalizes better.

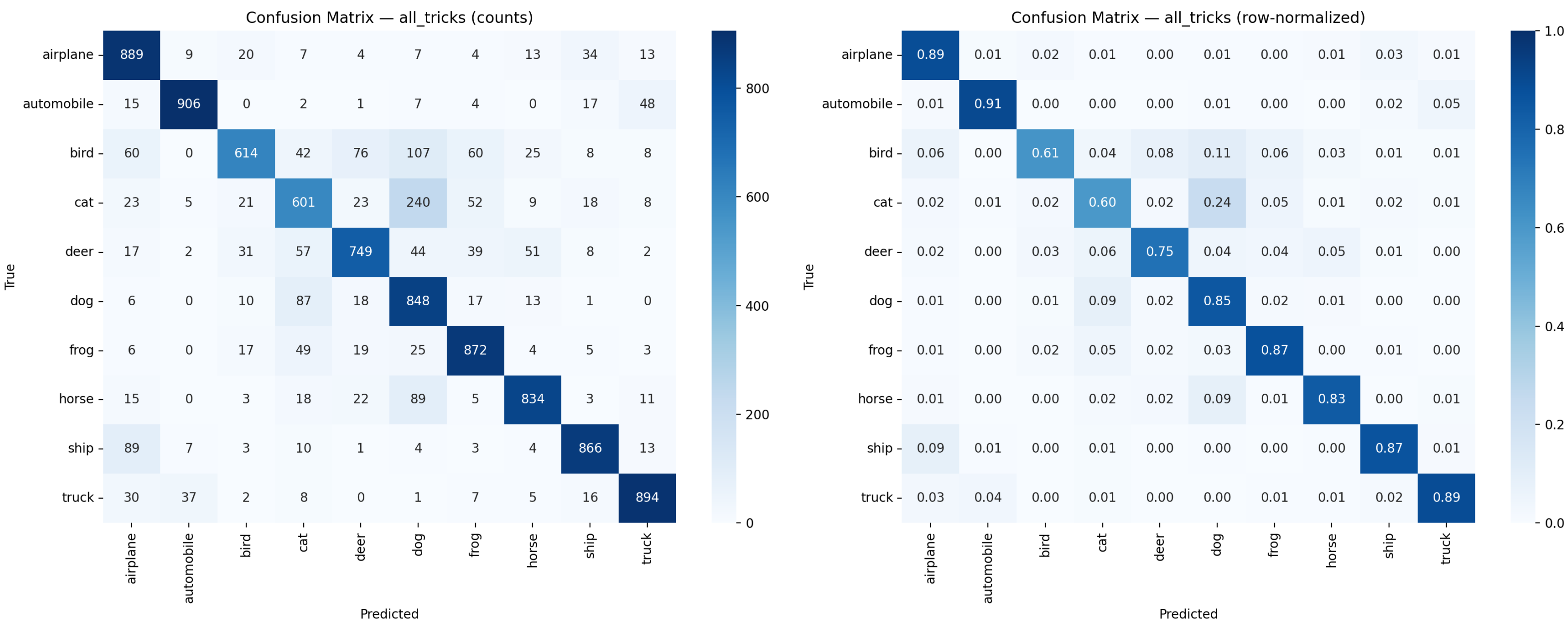

Figure 5: Confusion matrix for the all-tricks model. The strongest classes include automobile, airplane, truck, frog, ship, dog, and horse. The weakest classes are cat and bird, with the largest confusion being cat predicted as dog.

The all-tricks confusion matrix shows that the model performed well on vehicles and several visually distinct animal classes. Automobile was classified correctly at about 91%, airplane and truck at about 89%, frog and ship at about 87%, dog at about 85%, and horse at about 83%. The weakest classes were cat at about 60% and bird at about 61%. The largest single confusion was cat predicted as dog: 240 of 1000 cat test images were classified as dog. Other notable confusions included bird predicted as dog, horse predicted as dog, ship predicted as airplane, and automobile predicted as truck.

6 Discussion

The results suggest that the most important training trick for this small CNN was data augmentation. Augmentation improved test accuracy by 8.02 percentage points and reduced the generalization gap from 25.08 percentage points to 2.83 percentage points. This indicates that the baseline model had enough capacity to memorize the training set, and augmentation helped by making memorization harder.

Batch normalization improved test accuracy but did not reduce overfitting as much as augmentation. Its training accuracy was almost the same as the baseline, and its gap remained large. This suggests that, in this experiment, batch normalization mainly helped optimization rather than acting as a strong regularizer.

Dropout produced a similar accuracy improvement to batch normalization and lowered the train-test gap more noticeably. However, dropout still did not match the improvement from data augmentation. This may be because CIFAR-10 generalization depends heavily on spatial and visual variation, which image augmentation targets directly.

Label smoothing had the smallest accuracy improvement. A likely explanation is that label smoothing mainly affects confidence and calibration, while this project only evaluated accuracy and loss. A future version should include calibration metrics to better measure whether label smoothing improved the quality of predicted probabilities.

The all-tricks model performed strongly, but it did not beat augmentation alone. This is an important result: combining good techniques does not guarantee additive gains. Some techniques interact with each other, and the best hyperparameters for one method may not be optimal when all methods are used together. For example, a model trained with augmentation, dropout, and label smoothing at the same time may need more epochs, a different learning-rate schedule, or a different dropout probability.

Overall, this ablation study shows that the training pipeline matters substantially. The architecture was kept simple, but changing the training strategy moved test accuracy from 73.61% to above 81%. That improvement is large enough to be meaningful for a small CNN project and makes the work more substantial than simply training a classifier.

7 Limitations

This study has several limitations. First, the results were based on one random seed. A stronger experiment would run each configuration with seeds 0, 1, and 2 and report the mean and standard deviation. Without multi-seed evaluation, small differences such as 81.63% versus 81.32% should not be treated as statistically meaningful.

Second, the model was intentionally small. This made the experiments easier to run under limited compute, but the results may not transfer directly to larger architectures such as ResNet, DenseNet, or modern vision transformers.

Third, the hyperparameters were not extensively tuned. Every experiment used the same learning rate, optimizer, number of epochs, and batch size. This makes the comparison controlled, but it may disadvantage techniques that work best with different settings.

Fourth, the training accuracy for augmentation and all-tricks experiments should be interpreted carefully. If training accuracy is measured during training mode with random augmentation or dropout active, it may be lower than a clean evaluation of the same training set in evaluation mode. This explains why the all-tricks model had a negative train-test gap.

Finally, the study used accuracy, loss, and a confusion matrix, but it did not evaluate calibration, per-class precision and recall, training speed per epoch, or robustness to distribution shift. These metrics would make the analysis more complete.

8 Conclusion

This project evaluated the effect of common training techniques on a small CNN trained on CIFAR-10. The baseline model overfit strongly, reaching high training accuracy but only 73.61% test accuracy. Data augmentation produced the largest individual gain, improving test accuracy to 81.63% and greatly reducing the train-test gap. Batch normalization and dropout provided smaller but still positive improvements, while label smoothing had the smallest top-1 accuracy gain.

The all-tricks model achieved 81.32% test accuracy and showed strong performance in the confusion matrix, but it did not surpass augmentation alone in this single-seed run. The main takeaway is that simple training choices can significantly affect generalization even when the architecture is unchanged. For this small CNN on CIFAR-10, data augmentation was the most important single technique under the tested settings.

References

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